Bilateral Tensor Ring Decomposition for Thick Cloud Removal in Multitemporal Remote Sensing Images

Yu-Bang Zheng, *Member, IEEE*, Jia-Le Ma, Heng-Chao Li, *Senior Member, IEEE*, Zhi-Wei Shi, and Qing Zhu

***Abstract*—Cloud removal is crucial for enhancing the quality of remote sensing images (RSIs) and broadening their applicability. Tensor decomposition, which extracts latent correlations across multidimensions, has driven the development of various methods for thick cloud removal in multitemporal RSI (MTRSI). However, current tensor decomposition methods are not specifically tailored for MTRSIs, resulting in insufficient correlation representation and requiring high computational costs in processing MTRSIs. In this article, we construct a novel bilateral tensor ring (BTR) decomposition, the first method specifically designed for MTRSIs, which enables a customized representation of spatial, spectral, and temporal correlations with lower computational complexity. The fundamental idea behind BTR decomposition is to effectively distinguish between the weak spatial correlation and the strong spectral–temporal correlation while simultaneously capturing the interaction between these two components. With the support of BTR decomposition, we propose an MTRSI cloud removal model and develop an efficient proximal alternating minimization (PAM)-based algorithm to solve it. In theory, we prove a convergence guarantee for the algorithm. Extensive experimental results verify that our method offers superior cloud removal performance and delivers a *10- to 100-fold* acceleration in computational efficiency compared to the state-of-the-art tensor-based methods. The code is available at: https://yubangzheng.github.io.**

***Index Terms*—Bilateral tensor ring (BTR) decomposition, multitemporal remote sensing image (MTRSI), proximal alternating minimization (PAM), thick cloud removal.**

I. Introduction

THICK cloud cover presents a significant challenge in remote sensing image (RSI), causing substantial occlusion of surface information. This persistent issue greatly hinders the effective utilization of RSIs in critical applications, such as land cover classification, change detection, and disaster assessment [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20]. Therefore, developing efficient and robust cloud removal methods is of paramount importance to enhance the practical utility of RSIs. Traditional spatial- or spectral-based cloud removal methods often rely on single-temporal image processing technologies [21], [22], [23], [24]. However, reconstructing information in cloud-covered areas is challenging with images from only a single-temporal node because all spectral bands captured at the same time are uniformly affected by cloud interference, limiting their ability to provide clear and accurate surface details.

Multitemporal RSIs (MTRSIs) have become a powerful resource for thick cloud removal, thanks to their ability to capture sequential observations of the same region over time. This temporal redundancy enables the utilization of clear-sky conditions at different temporal nodes to reconstruct surface features obscured by clouds. By leveraging the rich spatial, spectral, and temporal information inherent in MTRSIs, numerous cloud removal methods have been developed. These methods have shown promising results in addressing the challenges of cloud removal, as demonstrated by a variety of studies [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38].

The core challenge for MTRSI cloud removal is how to effectively integrate and utilize the abundant spatial, spectral, and temporal information inherent in MTRSIs. The tensor, as a natural generalization of the matrix, is a powerful mathematical tool for representing MTRSIs. By decomposing the higher order tensor into a set of smaller scale factors, tensor decomposition enables the extraction of latent patterns and correlations across multidimensions [39], [40]. Therefore, many tensor decomposition-based methods for MTRSI cloud removal have been proposed [29], [30], [31], [32], [33], [34], [35], [36], [37], [38]. For example, Ji et al. [30] proposed a nonlocal Tucker decomposition-based method for MTRSI cloud removal, which reconstructs missing information by leveraging nonlocal correlations in the spatial domain and promoting low rankness in the fourth-order tensor groups across spatial, spectral, and temporal domains. He et al. [31] introduced a tensor ring (TR) decomposition-based method for MTRSI cloud removal to reconstruct missing information, incorporating total variation regularization to enhance the

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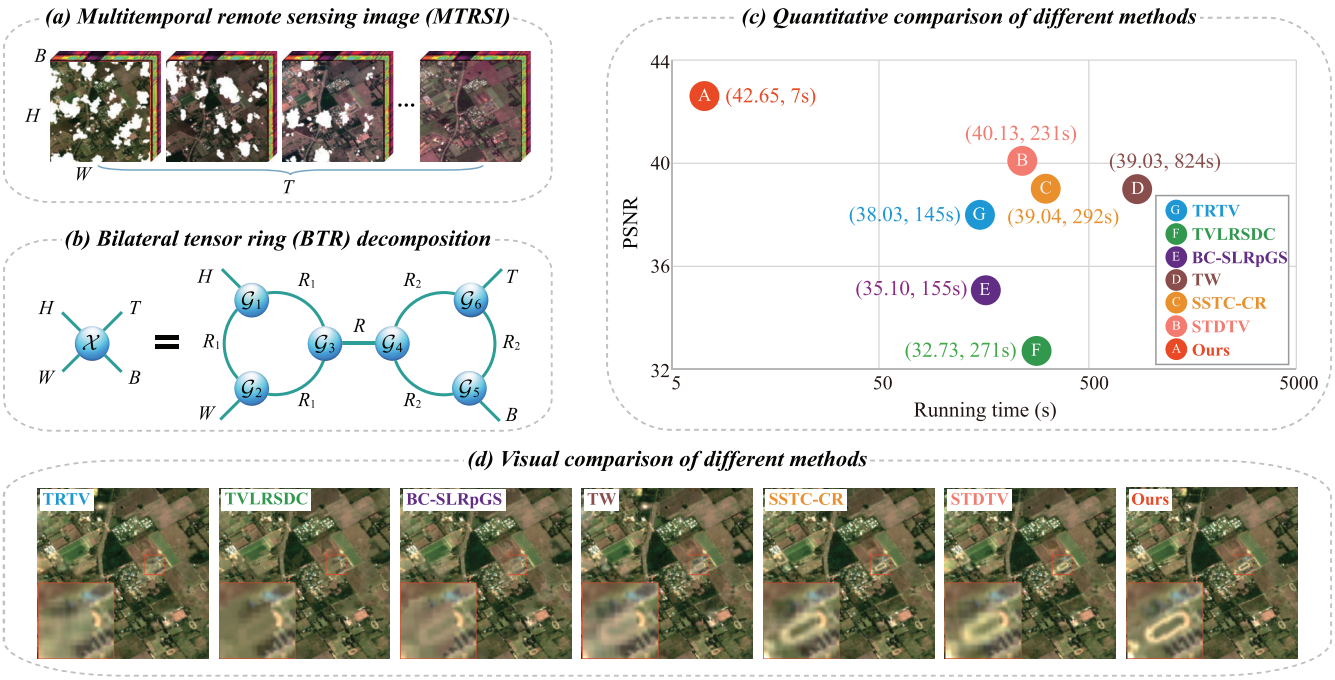


Fig. 1. (a) Cloud-contaminated MTRSI. (b) Graphical illustration of the proposed BTR decomposition. (c) Quantitative comparison of PSNR (↑) and running time (↓) for different methods under the experimental setup described in Section IV-B, with results representing mean values across four images in the *America* region. (d) Reconstructed MTRSIs (shown in RGB) of the second image in the *America* region at temporal node 1 obtained by different methods. The comparison methods, TRTV [31], TVLRSDC [32], BC-SLRpGS [33], TW [34], SSTC-CR [35], and STDTV [36], are derived from *TGRS'19*, *ISPRS'19*, *TGRS'22*, *NeurIPS'22*, *ISPRS'23*, and *TGRS'24*, respectively.

smoothness of spatial information. Zheng et al. [35] proposed a fully connected tensor network (FCTN) decomposition-based method for MTRSI cloud removal, which leverages the powerful representational capability of FCTN decomposition to achieve excellent reconstruction results. Zhang et al. [37] suggested a tensor singular value decomposition (SVD)-based method for MTRSI cloud removal, integrating deep spatiotemporal features from a 3-D convolutional neural network to enhance the reconstruction quality. By exploiting the multidimensional structure of MTRSIs, these methods can extract intricate spatial, spectral, and temporal information, thereby achieving promising results in cloud removal.

While tensor decomposition methods show great potential for MTRSI cloud removal, challenges remain. To be specific, existing methods rely on general-purpose tensor decomposition methods designed for high-dimensional data, rather than being specifically tailored for MTRSIs. This inevitably hampers their ability to fully and efficiently capture the abundant spatial, spectral, and temporal information. For instance, TR decomposition [31], [38], [41] struggles to fully capture the rich information in MTRSIs due to its intrinsic ring topology. In contrast, although FCTN decomposition [35], [42] excels in information capture, it incurs computational overhead due to its complex topology. This fact raises the question of how to design a tensor decomposition method specifically tailored to the unique characteristics of MTRSIs, and what the fundamental mechanism is that underpins its workings.

In this article, we construct a novel bilateral TR (BTR) decomposition [see Fig. 1(b)] that ingeniously crafts the design of two interconnected TR decompositions, allowing for a customized and compact representation of spatial, spectral, and temporal information of MTRSIs. Specifically, the method partitions four dimensions of MTRSIs into two components: one component comprises two spatial dimensions, while the other encompasses the spectral and temporal dimensions. The interaction between two TR decompositions captures the correlations across these components, while each decomposition independently models the internal correlations within its respective component. In terms of correlation representation, BTR decomposition not only distinguishes between weak spatial correlations and strong spectral–temporal correlations but also concurrently captures the interdependencies between these two components, thereby comprehensively leveraging the unique characteristics of MTRSIs. In terms of efficiency, for an MTRSI of size $H \times W \times B \times T$, the computational complexity of BTR decomposition is $\mathcal{O}(H^4R+H^2R^4)$, which is significantly lower than that of the advanced FCTN decomposition, whose complexity scales as $\mathcal{O}(H^4R^3+H^3R^6)$. Here, we simply assume that $H=W\geq B=T$, and all ranks involved in BTR and FCTN decompositions are set to R.

Based on BTR decomposition, we propose an MTRSI cloud removal model that can effectively remove cloud components while ensuring the preservation of fine-grained image details. In particular, thanks to the low complexity of BTR decomposition, the model achieves exceptional computational efficiency, enabling it to effectively process large-scale MTRSIs with remarkable speed and accuracy. To solve the proposed model, we develop an efficient proximal alternating minimization (PAM)-based algorithm and prove its theoretical convergence.

In summary, we make the following three contributions.

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1) We construct a novel BTR decomposition, the first tensor decomposition specifically tailored for MTRSIs, enabling a customized representation of spatial, spectral, and temporal correlations with reduced computational and storage complexity.
2) We propose an MTRSI cloud removal model based on BTR decomposition and develop a PAM-based algorithm to solve it. Theoretically, we prove a convergence guarantee for the proposed method.
3) Extensive experimental results verify numerically that our method achieves superior cloud removal performance and provides a *10- to 100-fold* acceleration in computational efficiency compared to the state-of-the-art tensor-based methods (see Fig. 1(c) and (d) as an example).

A. Related Works

We review related MTRSI thick cloud removal methods. Based on the availability of cloud location (also called cloud mask) information, existing MTRSI thick cloud removal methods can be categorized into three main types: mask-unknown, mask-known, and mask-semi-known methods.

Mask-unknown methods do not rely on cloud location information, offering the advantage of being applicable even when cloud mask is unavailable [32], [33], [43], [44], [45], [46]. Within this framework, Chen et al. [32] proposed an MTRSI cloud removal method by decomposing the degraded image into low-rank (target image) and sparse (cloud) components while introducing total variation regularization to enhance the continuity of the cloud component. Subsequently, Ji et al. [33] suggested a box-constrained smooth low-rank plus group sparse method for MTRSI cloud removal, employing tensor SVD to characterize the global low rankness of the image component and using a group sparse function to constrain the cloud component. Yang et al. [44] designed a low-rank tensor factorization framework with factor-based group sparsity regularization to capture the spectral–temporal consistency prior of MTRSIs and proposed an MTRSI cloud removal method, achieving superior performance with low computational complexity. Chen et al. [46] presented a cloud removal method that integrates model-driven low-rank regularization with a self-supervised data-driven network. However, the above cloud removal methods usually result in reduced accuracy in cloud-free regions due to the lack of guidance of cloud masks. Therefore, the above methods typically incorporate cloud detection techniques after the cloud removal process and improve the accuracy of cloud-free regions through projection operations.

Mask-known methods leverage cloud location information to accurately guide the cloud removal process. Within this framework, early methods primarily leveraged temporal complementarity to facilitate cloud removal, and these methods can be broadly categorized into three types: temporal replacement [47], [48], [49], temporal filtering [50], [51], and temporal learning [52], [53]. However, the effectiveness of these methods is limited by their reliance on the assumption of minimal scene change over time. Significant variations in the scene, such as seasonal changes and land use modifications, can lead to inaccuracies in cloud-free pixel estimation. To address this issue, recent cloud removal methods have integrated spatial, spectral, and temporal information of MTRSIs. For example, Chen et al. [54] proposed a method that combines cloud-free Landsat data with a spatiotemporal weighted regression model and reduces biases using a prior modification term. Zhang et al. [55] introduced a spatiotemporal patch group deep learning framework for cloud removal, incorporating a global–local loss function and weighted aggregation to enhance reconstruction accuracy. Darbaghshahi et al. [56] suggested a dual-generative adversarial network (GAN) for MTRSI cloud removal, incorporating dilated residual inception blocks to enhance feature extraction and demonstrating superior performance on MTRSIs. Since the tensor is a powerful mathematical tool for representing MTRSIs, mask-known cloud removal methods can be regarded as a low-rank tensor completion problem with structural information missing. Therefore, many methods based on tensor decomposition [35], [38], [57], [58], [59], as well as joint tensor decomposition and deep regularization techniques, have been applied to the cloud removal problem. For example, Wang et al. [60] proposed an unsupervised domain factorization network for MTRSI cloud removal, which leverages untrained neural networks to extract spatial and spectral factors from each time frame and introduces a low-rank spatial factor module to jointly model spatial information across time. Peng et al. [61] proposed a deep domain fidelity method that incorporates low-rank regularization based on TR decomposition, aiming to enhance cloud removal performance and preserve fine details in MTRSIs. These methods leverage the capabilities of tensor decomposition to effectively model the spatial, spectral, and temporal correlations in MTRSIs, while deep regularization techniques are employed to enforce smoothness and consistency in the reconstructed data [60], [61]. By utilizing the cloud location information, the above cloud removal methods can significantly improve the performance. However, the main limitation of these methods is their dependency on the availability of accurate cloud masks. If the cloud mask is inaccurate, it can negatively affect the cloud removal process, leading to poor recovery of the missing information.

Mask-semi-known methods integrate aspects of both mask-known and mask-unknown methods. They utilize partial cloud location information but do not require fully accurate cloud masks [35], [62]. This balance enables them to remain robust under conditions with inaccurate cloud masks while preserving reasonable accuracy in cloud-free areas. Within this framework, Lin et al. [62] proposed a robust cloud removal method that addresses the issue of inaccurate cloud masks by using coupled tensor factorization to exploit the relationship between the abundances of MTRSIs in the same scene. Subsequently, Zheng et al. [35] introduced an FCTN decomposition-based method for MTRSI cloud removal, which leverages the powerful representational capability of FCTN decomposition to capture the spatial, spectral, and temporal information of MTRSIs and achieves excellent cloud removal results. However, the complex topology of FCTN decomposition makes this method highly time-consuming.

Our method belongs to mask-semi-known methods within the tensor decomposition family. Different from existing meth-

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ods, we construct the first tensor decomposition specifically designed for MTRSIs: the BTR decomposition. This decomposition allows for a tailored representation of spatial, spectral, and temporal correlations while simultaneously reducing both computational and storage complexity. In our method, there is no need for complex regularization; the newly constructed BTR decomposition alone is sufficient to achieve superior results compared to existing methods, while significantly reducing computational time. Not only that, the method also allows for the integration of advanced regularization terms, providing opportunities for further performance enhancement.

B. Outlines

The remainder of this article is organized as follows. Section II introduces the notations and provides preliminary knowledge. Section III constructs BTR decomposition, presents the associated MTRSI cloud removal model, and develops a PAM-based solving algorithm with a theoretical convergence guarantee. Section IV conducts extensive experiments to evaluate the performance of the proposed method. Finally, Section V concludes this article.

II. Notations and Preliminaries

We use bold lowercase letters, e.g., $\mathbf{x}$, bold uppercase letters, e.g., $\mathbf{X}$, and calligraphic uppercase letters, e.g., $\mathcal{X}$, to denote vectors, matrices, and tensors, respectively. For simplicity, we denote the ordered set $\{x_1, x_2, \ldots, x_n\}$ by $x_{1:n}$.

The *Frobenius norm* of a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ is defined as $\|\mathcal{X}\|_F = \left(\sum_{i_1,i_2,\ldots,i_N} |\mathcal{X}(i_1, i_2, \ldots, i_N)|^2\right)^{1/2}$ [39].

The *unfolding*, also called *matricization*, is an operation that restructures a tensor into a matrix by mapping a designated subset of its dimensions to the matrix rows and projecting the remaining dimensions into the matrix columns. For instance, an $I_1 \times I_2 \times I_3 \times I_4$ tensor can be restructured into an $I_1 \times I_2 I_3 I_4$ matrix, represented as $\mathbf{X}_{[1|2,3,4]} = \text{Unfold}(\mathcal{X}, [1|2,3,4])$. It can also be restructured into an $I_1 I_2 \times I_3 I_4$ matrix, denoted by $\mathbf{X}_{[1,2|3,4]} = \text{Unfold}(\mathcal{X}, [1,2|3,4])$.[1]

Given two tensors, the *tensor contraction* between them produces a new tensor by pairing specific dimensions, multiplying the corresponding elements, and summing over the paired indices. For instance, given a third-order tensor $\mathcal{X} \in \mathbb{R}^{I \times J \times K}$ and fourth-order tensor $\mathcal{Y} \in \mathbb{R}^{M \times I \times N \times K}$, the tensor contraction between the 1st and 3rd dimensions of $\mathcal{X}$ and the 2nd and 4th dimensions of $\mathcal{Y}$ yields a tensor $\mathcal{Z} = \mathcal{X} \times_{1,3}^{2,4} \mathcal{Y} \in \mathbb{R}^{J \times M \times N}$, whose elements are calculated as

$$\mathcal{Z}(j, m, n) = \sum_{i=1}^{I} \sum_{k=1}^{K} \mathcal{X}(i, j, k)\, \mathcal{Y}(m, i, n, k).$$

For more details on tensor contraction, please refer to [63].

Theorem 1 [63]: Let tensors $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times, \ldots, \times I_N}$ and $\mathcal{Y} \in \mathbb{R}^{J_1 \times J_2 \times, \ldots, \times J_M}$; then, we have

$$\mathcal{Z} = \mathcal{X} \times_{n_{1:d}}^{m_{1:d}} \mathcal{Y} \Leftrightarrow$$
$$\mathbf{Z}_{[1:N-d|N-d+1:N+M-2d]} = \mathbf{X}_{[n_{d+1:N}|n_{1:d}]} \mathbf{Y}_{[m_{1:d}|m_{d+1:M}]}$$

where $\mathbf{n}$ and $\mathbf{m}$ are arbitrary rearrangements of $(1, 2, \ldots, N)$ and $(1, 2, \ldots, M)$, respectively.

Theorem 1 implies that tensor contraction is computationally equivalent to matrix multiplication. For example, given two tensors $\mathcal{X} \in \mathbb{R}^{I \times J \times K}$ and $\mathcal{Y} \in \mathbb{R}^{M \times I \times N \times K}$, the contraction $\mathcal{Z} = \mathcal{X} \times_{1,3}^{2,4} \mathcal{Y} \in \mathbb{R}^{J \times M \times N}$ is equivalent to the matrix multiplication $\mathbf{Z}_{[1|2,3]} = \mathbf{X}_{[2|1,3]} \mathbf{Y}_{[2,4|1,3]} \in \mathbb{R}^{J \times MN}$.

TR decomposition factorizes a 3rd-order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ as

$$\mathcal{X} = \mathcal{G}_1 \times_3^1 \mathcal{G}_2 \times_{4,1}^{1,3} \mathcal{G}_3 \tag{1}$$

where factors $\mathcal{G}_1 \in \mathbb{R}^{R_1 \times I_1 \times R_2}, \mathcal{G}_2 \in \mathbb{R}^{R_2 \times I_2 \times R_3}$, and $\mathcal{G}_3 \in \mathbb{R}^{R_3 \times I_3 \times R_1}$ [41]. We denote (1) by $\mathcal{X} = \text{TR}(\{\mathcal{G}_k\}_{k=1}^3)$.

Not limited to TR decomposition, other tensor decomposition methods have also been explored for the MTRSI cloud removal problem, including Tucker decomposition, tensor SVD, and FCTN decomposition [29], [30], [31], [35], [36], [37], [38]. However, existing tensor decomposition methods are primarily designed as general-purpose frameworks for high-dimensional data, rather than being explicitly tailored to the unique characteristics of MTRSIs. This limits their effectiveness and efficiency in capturing the spatial, spectral, and temporal correlations in MTRSIs, reducing the reconstruction capability and computational efficiency in MTRSI cloud removal.

III. Proposed Model and Solving Algorithm

In this section, we first construct a BTR decomposition, a novel method tailored for MTRSIs. We then present the associated MTRSI cloud removal model and develop a PAM-based solving algorithm with a theoretical convergence guarantee.

A. BTR Decomposition

We begin with the definition of the following BTR decomposition.

Definition 1 (BTR Decomposition): Let $\mathcal{X} \in \mathbb{R}^{H \times W \times B \times T}$ be a fourth-order tensor such that

$$\begin{aligned}\mathcal{X} &= [\text{TR}(\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3)] \times_3^1 [\text{TR}(\mathcal{G}_4, \mathcal{G}_5, \mathcal{G}_6)] \\ &= \left[\mathcal{G}_1 \times_3^1 \mathcal{G}_2 \times_{4,1}^{1,3} \mathcal{G}_3\right] \times_3^1 \left[\mathcal{G}_4 \times_3^1 \mathcal{G}_5 \times_{4,1}^{1,3} \mathcal{G}_6\right]\end{aligned} \tag{2}$$

where $\mathcal{G}_1 \in \mathbb{R}^{R_1 \times H \times R_1}, \mathcal{G}_2 \in \mathbb{R}^{R_1 \times W \times R_1}$, $\mathcal{G}_3 \in \mathbb{R}^{R_1 \times R \times R_1}, \mathcal{G}_4 \in \mathbb{R}^{R_2 \times R \times R_2}, \mathcal{G}_5 \in \mathbb{R}^{R_2 \times B \times R_2}$, and $\mathcal{G}_6 \in \mathbb{R}^{R_2 \times T \times R_2}$ are called BTR factors. Then, we call (2) a BTR decomposition of $\mathcal{X}$ and simply denote it by $\mathcal{X} = \text{BTR}(\{\mathcal{G}_k\}_{k=1}^6)$.

The proposed BTR decomposition is a method explicitly tailored to the unique characteristics of MTRSIs, enabling a customized representation of spatial, spectral, and temporal correlations with reduced storage and computational complexity. In Remark 1, we explain the fundamental mechanism of BTR decomposition.

Remark 1 (Correlation Representation): Given an MTRSI $\mathcal{X} \in \mathbb{R}^{H \times W \times B \times T}$ (H: spatial heights, W: spatial widths, B: spectral bands, and T: temporal nodes), BTR decomposition partitions its dimensions into two components: one comprises two spatial dimensions and the other includes the spectral–temporal dimensions [see Fig. 1(b)].

1) The interaction between two TR decompositions captures the correlations between the spatial dimensions

[1] Unfold($\mathcal{X}, [1,2|3,4]$) and Unfold($\mathcal{X}, [2,1|3,4]$) produce matrices of the same size, but the arrangement of elements within the matrices follows different rules. Readers interested in the details can refer to [63].

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and the spectral–temporal dimensions, with the strength controlled by the rank parameter R.

2) The left TR decomposition captures the correlations within the spatial dimensions, with the strength controlled by rank parameter R_1.
3) The right TR decomposition captures the correlations within the spectral–temporal dimensions, with the strength controlled by rank parameter R_2.

In summary, BTR decomposition not only distinguishes between the weak spatial correlation and the strong spectral–temporal correlations but also concurrently captures the interdependencies between these two components.

In Remark 2, we compare the complexities of BTR decomposition and the advanced FCTN decomposition.

Remark 2 (Storage and Computational Complexity): Given an MTRSI $\mathcal{X} \in \mathbb{R}^{H\times W\times B\times T}$, the storage complexity of the proposed BTR decomposition is $\mathcal{O}((H+W+B+T)R^2)$, which is lower than that of the advanced FCTN decomposition, whose storage complexity is $\mathcal{O}((H+W+B+T)R^3)$. Simply assuming that $H=W\geq B=T$, the computational complexity of BTR decomposition is $\mathcal{O}(H^4R+H^2R^4)$, which is significantly lower than that of the advanced FCTN decomposition, whose complexity scales as $\mathcal{O}(H^4R^3+H^3R^6)$.

B. BTR Decomposition for MTRSI Cloud Removal

With the support of BTR decomposition, we propose a novel MTRSI cloud removal model. Given a cloud-contaminated observation $\mathcal{Y} \in \mathbb{R}^{H\times W\times B\times T}$, the BTR decomposition-based MTRSI cloud removal model can be formulated as

$$\min_{\mathcal{G},\mathcal{X}} \frac{1}{2}\|\mathcal{M}\odot(\mathcal{Y}-\mathcal{X})\|_F^2+\frac{\alpha}{2}\|\mathcal{X}-\text{BTR}\left(\{\mathcal{G}_k\}_{k=1}^6\right)\|_F^2 \tag{3}$$

where $\odot$ is the componentwise multiplication; $\mathcal{X} \in \mathbb{R}^{H\times W\times B\times T}$ is the target MTRSI; $\mathcal{M} \in \mathbb{R}^{H\times W\times B\times T}$ is the cloud mask, a binary tensor in which ones point to cloud-free pixels and zeros point to cloud-contaminated pixels; $\mathcal{G}_k$ for $k = 1, 2, \ldots, 6$ is BTR factors; $\mathcal{G} = \{\mathcal{G}_k\}_{k=1}^6$; and $\alpha > 0$ is a parameter.

Based on Definition 1, we introduce two auxiliary variables $\mathcal{A} \in \mathbb{R}^{H\times W\times R}$ and $\mathcal{B} \in \mathbb{R}^{R\times B\times T}$ and rewrite (3) as

$$\begin{aligned}&\min_{\mathcal{A},\mathcal{B},\mathcal{G},\mathcal{X}} \frac{1}{2}\|\mathcal{M}\odot(\mathcal{Y}-\mathcal{X})\|_F^2+\frac{\alpha}{2}\|\mathcal{X}-\mathcal{A}\times_3^1\mathcal{B}\|_F^2\\ &\text{s.t.}\quad \mathcal{A}=\text{TR}\left(\{\mathcal{G}_k\}_{k=1}^3\right),\ \mathcal{B}=\text{TR}\left(\{\mathcal{G}_k\}_{k=4}^6\right).\end{aligned} \tag{4}$$

By introducing the penalty terms $(\beta_1/2)\|\mathcal{A}-\text{TR}(\{\mathcal{G}_k\}_{k=1}^3)\|_F^2$ and $(\beta_2/2)\|\mathcal{B}-\text{TR}(\{\mathcal{G}_k\}_{k=4}^6)\|_F^2$, problem (4) can be rewritten as the following unconstrained problem:

$$\begin{aligned}&\min_{\mathcal{A},\mathcal{B},\mathcal{G},\mathcal{X}} \frac{1}{2}\|\mathcal{M}\odot(\mathcal{Y}-\mathcal{X})\|_F^2+\frac{\alpha}{2}\|\mathcal{X}-\mathcal{A}\times_3^1\mathcal{B}\|_F^2\\ &+\frac{\beta_1}{2}\|\mathcal{A}-\text{TR}\left(\{\mathcal{G}_k\}_{k=1}^3\right)\|_F^2\\ &+\frac{\beta_2}{2}\|\mathcal{B}-\text{TR}\left(\{\mathcal{G}_k\}_{k=4}^6\right)\|_F^2\end{aligned} \tag{5}$$

where $\beta_1 > 0$ and $\beta_2 > 0$ are penalty parameters.

Next, we use PAM to solve (5), whose solution can be optimized by alternately updating the following set of equations:

$$\begin{cases}\mathcal{A}=\underset{\mathcal{A}}{\text{argmin}}\, f(\mathcal{A},\mathcal{B},\mathcal{G},\mathcal{X})+\frac{\rho}{2}\|\mathcal{A}-\hat{\mathcal{A}}\|_F^2\\ \mathcal{B}=\underset{\mathcal{B}}{\text{argmin}}\, f(\mathcal{A},\mathcal{B},\mathcal{G},\mathcal{X})+\frac{\rho}{2}\|\mathcal{B}-\hat{\mathcal{B}}\|_F^2\\ \mathcal{G}_k=\underset{\mathcal{G}_k}{\text{argmin}}\, f(\mathcal{A},\mathcal{B},\mathcal{G},\mathcal{X})+\frac{\rho}{2}\|\mathcal{G}_k-\hat{\mathcal{G}}_k\|_F^2\\ \mathcal{X}=\underset{\mathcal{X}}{\text{argmin}}\, f(\mathcal{A},\mathcal{B},\mathcal{G},\mathcal{X})+\frac{\rho}{2}\|\mathcal{X}-\hat{\mathcal{X}}\|_F^2\end{cases} \tag{6}$$

where $f(\mathcal{A},\mathcal{B},\mathcal{G},\mathcal{X})$ denotes the objective function of (5); $k = 1, 2, \ldots, 6, \rho > 0$ is a proximal parameter; and $\hat{(\cdot)}$ is the result at the previous iteration of the corresponding subproblem.

1) Update $\mathcal{A}$: The $\mathcal{A}$-subproblem can be written as

$$\begin{aligned}\underset{\mathcal{A}}{\text{argmin}}\ &\frac{\alpha}{2}\|\mathcal{X}-\mathcal{A}\times_3^1\mathcal{B}\|_F^2+\frac{\beta_1}{2}\|\mathcal{A}-\text{TR}\left(\{\mathcal{G}_k\}_{k=1}^3\right)\|_F^2\\ &+\frac{\rho}{2}\|\mathcal{A}-\hat{\mathcal{A}}\|_F^2.\end{aligned} \tag{7}$$

According to Theorem 1, (7) can be rewritten in the following matrix form:

$$\underset{\mathbf{A}}{\text{argmin}}\ \frac{\alpha}{2}\|\mathbf{X}-\mathbf{A}\mathbf{B}\|_F^2+\frac{\beta_1}{2}\|\mathbf{A}-\mathbf{C}\|_F^2+\frac{\rho}{2}\|\mathbf{A}-\hat{\mathbf{A}}\|_F^2 \tag{8}$$

with

$$\begin{cases}\mathbf{X}=\text{Unfold}(\mathcal{X},[1,2|3,4])\in\mathbb{R}^{HW\times BT}\\ \mathbf{A}=\text{Unfold}(\mathcal{A},[1,2|3])\in\mathbb{R}^{HW\times R}\\ \mathbf{B}=\text{Unfold}(\mathcal{B},[1|2,3])\in\mathbb{R}^{R\times BT}\\ \mathbf{C}=\text{Unfold}(\text{TR}(\{\mathcal{G}_k\}_{k=1}^3),[1,2|3])\in\mathbb{R}^{HW\times R}.\end{cases}$$

The objective function of (8) is differentiable, and thus, we can obtain the solution as

$$\mathbf{A}=\left(\alpha\mathbf{X}\mathbf{B}^{\text{T}}+\beta_1\mathbf{C}+\rho\hat{\mathbf{A}}\right)\left(\alpha\mathbf{B}\mathbf{B}^{\text{T}}+(\beta_1+\rho)\mathbf{I}\right)^{-1}. \tag{9}$$

The cost of computing $\mathbf{A}$ is $\mathcal{O}(HWBTR)$.

2) Update $\mathcal{B}$: The $\mathcal{B}$-subproblem can be written as

$$\begin{aligned}\underset{\mathcal{B}}{\text{argmin}}\ &\frac{\alpha}{2}\|\mathcal{X}-\mathcal{A}\times_3^1\mathcal{B}\|_F^2+\frac{\beta_2}{2}\|\mathcal{B}-\text{TR}\left(\{\mathcal{G}_k\}_{k=4}^6\right)\|_F^2\\ &+\frac{\rho}{2}\|\mathcal{B}-\hat{\mathcal{B}}\|_F^2.\end{aligned} \tag{10}$$

According to Theorem 1, (10) can be rewritten as the following matrix form:

$$\underset{\mathbf{B}}{\text{argmin}}\ \frac{\alpha}{2}\|\mathbf{X}-\mathbf{A}\mathbf{B}\|_F^2+\frac{\beta_2}{2}\|\mathbf{B}-\mathbf{D}\|_F^2+\frac{\rho}{2}\|\mathbf{B}-\hat{\mathbf{B}}\|_F^2 \tag{11}$$

where matrix $\mathbf{D} = \text{Unfold}(\text{TR}(\{\mathcal{G}_k\}_{k=4}^6), [1|2, 3]) \in \mathbb{R}^{R\times BT}$. The objective function of (11) is also differentiable, and thus we can obtain the solution as

$$\mathbf{B}=\left(\alpha\mathbf{A}^{\text{T}}\mathbf{A}+(\beta_2+\rho)\mathbf{I}\right)^{-1}\left(\alpha\mathbf{A}^{\text{T}}\mathbf{X}+\beta_2\mathbf{D}+\rho\hat{\mathbf{B}}\right). \tag{12}$$

The cost of computing $\mathbf{B}$ is $\mathcal{O}(HWBTR)$.

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3) Update $\mathcal{G}$: For $k = 1, 2, 3$, the $\mathcal{G}_k$-subproblem can be written as

$$\underset{\mathcal{G}_k}{\operatorname{argmin}} \frac{\beta_1}{2}\|\mathcal{A} - \mathrm{TR}\left(\{\mathcal{G}_k\}_{k=1}^{3}\right)\|_F^2 + \frac{\rho}{2}\|\mathcal{G}_k - \hat{\mathcal{G}}_k\|_F^2. \tag{13}$$

According to Theorem 1, (13) can be rewritten as the following matrix form:

$$\underset{\mathbf{G}_k}{\operatorname{argmin}} \frac{\beta_1}{2}\|\mathbf{A}_k - \mathbf{G}_k\mathbf{M}_k\|_F^2 + \frac{\rho}{2}\|\mathbf{G}_k - \hat{\mathbf{G}}_k\|_F^2 \tag{14}$$

with

$$\begin{cases} \mathbf{A}_1 = \mathrm{Unfold}(\mathcal{A}, [1|2,3]) \in \mathbb{R}^{H\times WR} \\ \mathbf{G}_1 = \mathrm{Unfold}(\mathcal{G}_1, [2|3,1]) \in \mathbb{R}^{H\times R_1^2} \\ \mathbf{M}_1 = \mathrm{Unfold}(\mathcal{G}_2 \times_3^1 \mathcal{G}_3, [1,4|2,3]) \in \mathbb{R}^{R_1^2\times WR} \end{cases}$$

$$\begin{cases} \mathbf{A}_2 = \mathrm{Unfold}(\mathcal{A}, [2|3,1]) \in \mathbb{R}^{W\times RH} \\ \mathbf{G}_2 = \mathrm{Unfold}(\mathcal{G}_2, [2|3,1]) \in \mathbb{R}^{W\times R_1^2} \\ \mathbf{M}_2 = \mathrm{Unfold}(\mathcal{G}_3 \times_3^1 \mathcal{G}_1, [1,4|2,3]) \in \mathbb{R}^{R_1^2\times RH} \end{cases}$$

and

$$\begin{cases} \mathbf{A}_3 = \mathrm{Unfold}(\mathcal{A}, [3|1,2]) \in \mathbb{R}^{R\times HW} \\ \mathbf{G}_3 = \mathrm{Unfold}(\mathcal{G}_3, [2|3,1]) \in \mathbb{R}^{R\times R_1^2} \\ \mathbf{M}_3 = \mathrm{Unfold}(\mathcal{G}_1 \times_3^1 \mathcal{G}_2, [1,4|2,3]) \in \mathbb{R}^{R_1^2\times HW}. \end{cases}$$

The objective function of (14) is also differentiable, and thus, we can obtain the solution as

$$\mathbf{G}_k = \left(\beta_1\mathbf{A}_k\mathbf{M}_k^{\mathrm{T}} + \rho\hat{\mathbf{G}}_k\right)\left(\beta_1\mathbf{M}_k\mathbf{M}_k^{\mathrm{T}} + \rho\mathbf{I}\right)^{-1}. \tag{15}$$

The costs of computing $\mathcal{G}_1$–$\mathcal{G}_3$ are $\mathcal{O}(WRR_1^4 + HWRR_1^2)$, $\mathcal{O}(HRR_1^4 + HWRR_1^2)$, and $\mathcal{O}(HWR_1^4 + HWRR_1^2)$, respectively.

For $k = 4, 5, 6$, the $\mathcal{G}_k$-subproblem can be written as

$$\underset{\mathcal{G}_k}{\operatorname{argmin}} \frac{\beta_2}{2}\|\mathcal{B} - \mathrm{TR}\left(\{\mathcal{G}_k\}_{k=4}^{6}\right)\|_F^2 + \frac{\rho}{2}\|\mathcal{G}_k - \hat{\mathcal{G}}_k\|_F^2. \tag{16}$$

According to Theorem 1, (16) can be rewritten in the following matrix form:

$$\underset{\mathbf{G}_k}{\operatorname{argmin}} \frac{\beta_2}{2}\|\mathbf{B}_k - \mathbf{G}_k\mathbf{M}_k\|_F^2 + \frac{\rho}{2}\|\mathbf{G}_k - \hat{\mathbf{G}}_k\|_F^2 \tag{17}$$

with

$$\begin{cases} \mathbf{B}_4 = \mathrm{Unfold}(\mathcal{B}, [1|2,3]) \in \mathbb{R}^{R\times BT} \\ \mathbf{G}_4 = \mathrm{Unfold}(\mathcal{G}_4, [2|3,1]) \in \mathbb{R}^{R\times R_2^2} \\ \mathbf{M}_4 = \mathrm{Unfold}(\mathcal{G}_5 \times_3^1 \mathcal{G}_6, [1,4|2,3]) \in \mathbb{R}^{R_2^2\times BT} \end{cases}$$

$$\begin{cases} \mathbf{B}_5 = \mathrm{Unfold}(\mathcal{B}, [2|3,1]) \in \mathbb{R}^{B\times TR} \\ \mathbf{G}_5 = \mathrm{Unfold}(\mathcal{G}_5, [2|3,1]) \in \mathbb{R}^{B\times R_2^2} \\ \mathbf{M}_5 = \mathrm{Unfold}(\mathcal{G}_6 \times_3^1 \mathcal{G}_4, [1,4|2,3]) \in \mathbb{R}^{R_2^2\times TR} \end{cases}$$

and

$$\begin{cases} \mathbf{B}_6 = \mathrm{Unfold}(\mathcal{B}, [3|1,2]) \in \mathbb{R}^{T\times RB} \\ \mathbf{G}_6 = \mathrm{Unfold}(\mathcal{G}_6, [2|3,1]) \in \mathbb{R}^{T\times R_2^2} \\ \mathbf{M}_6 = \mathrm{Unfold}(\mathcal{G}_4 \times_3^1 \mathcal{G}_5, [1,4|2,3]) \in \mathbb{R}^{R_2^2\times RB}. \end{cases}$$

The objective function of (17) is also differentiable, and thus, we can obtain the solution as

$$\mathbf{G}_k = \left(\beta_2\mathbf{B}_k\mathbf{M}_k^{\mathrm{T}} + \rho\hat{\mathbf{G}}_k\right)\left(\beta_2\mathbf{M}_k\mathbf{M}_k^{\mathrm{T}} + \rho\mathbf{I}\right)^{-1}. \tag{18}$$

The costs of computing $\mathcal{G}_4$–$\mathcal{G}_6$ are $\mathcal{O}(BTR_2^4 + BTRR_2^2)$, $\mathcal{O}(TRR_2^4 + BTRR_2^2)$, and $\mathcal{O}(BRR_2^4 + BTRR_2^2)$, respectively.

4) Update $\mathcal{X}$: The $\mathcal{X}$-subproblem can be written as

$$\underset{\mathcal{X}}{\operatorname{argmin}} \frac{1}{2}\|\mathcal{M}\odot(\mathcal{Y} - \mathcal{X})\|_F^2 + \frac{\alpha}{2}\|\mathcal{X} - \mathcal{A}\times_3^1 \mathcal{B}\|_F^2 + \frac{\rho}{2}\|\mathcal{X} - \hat{\mathcal{X}}\|_F^2. \tag{19}$$

The objective function of (19) is also differentiable, and thus, we can obtain the solution as

$$\mathcal{X} = \left(\mathcal{M}\odot\mathcal{Y} + \alpha\left(\mathcal{A}\times_3^1 \mathcal{B}\right) + \rho\hat{\mathcal{X}}\right) \oslash \left(\mathcal{M} + (\alpha + \rho)\,\mathcal{J}\right) \tag{20}$$

where $\oslash$ is the componentwise division and $\mathcal{J} \in \mathbb{R}^{H\times W\times B\times T}$ is a tensor whose elements are all equal to 1. The cost of computing $\mathcal{X}$ is $\mathcal{O}(HWBTR)$.

5) Mask Refinement Strategy ($\mathcal{M}$): We employ a progressive mask refinement strategy, iteratively enhancing the cloud mask at each step. With each iteration, the mask becomes progressively more accurate, thereby improving the cloud removal process. This iterative feedback loop guarantees that, even with an inaccurate initial mask, the developed algorithm consistently yields reliable results. Specifically, we update $\mathcal{M}$ through the following threshold function:

$$\mathcal{M}(h,w,b,t) = \begin{cases} 0, & \text{if } \mathcal{H}(h,w,t) \geq \omega_t \\ 1, & \text{if } \mathcal{H}(h,w,t) < \omega_t \end{cases} \tag{21}$$

where $\mathcal{H}(h,w,t) = 1/B\sum_{b=1}^{B}|\mathcal{Y}(h,w,b,t) - \mathcal{X}(h,w,b,t)|$, $\omega_t = \min\{\mathcal{H}(h,w,t)|(h,w,t) \in \Omega\}$ with the index set $\Omega = \{(h,w,t)|\mathcal{M}(h,w,1,t) = 0\}_{h=1,w=1,t=1}^{H,W,T}$. The cost of computing $\mathcal{M}$ is $\mathcal{O}(HWBT)$.

Algorithm 1 PAM-Based Algorithm for the Proposed Model

Input: The cloud-contaminated MTRSI $\mathcal{Y} \in \mathbb{R}^{H\times W\times B\times T}$, the initial mask $\mathcal{M}$, the ranks R, R_1, and R_2, and parameters $\beta_1 = \beta_2, \alpha = 1$, and $\rho = 0.01$.

Initialization: Let $\mathcal{X} = \mathcal{Y}$, and initialize $\mathcal{A}, \mathcal{B}$, and $\mathcal{G}_k$ for $k = 1, 2, \ldots, 6$ with all elements equal to 1.

for $i = 1$ to 100 **do**
  Let $\hat{\mathcal{X}} = \mathcal{X}$.
  Update $\mathbf{A}$ via (9) and reshape it to $\mathcal{A} \in \mathbb{R}^{H\times W\times R}$.
  Update $\mathbf{B}$ via (12) and reshape it to $\mathcal{B} \in \mathbb{R}^{R\times B\times T}$.
  Update $\mathbf{G}_1, \mathbf{G}_2$, and $\mathbf{G}_3$ via (15), and reshape them to $\mathcal{G}_1 \in \mathbb{R}^{R_1\times H\times R_1}$, $\mathcal{G}_2 \in \mathbb{R}^{R_1\times W\times R_1}$, and $\mathcal{G}_3 \in \mathbb{R}^{R_1\times R\times R_1}$.
  Update $\mathbf{G}_4, \mathbf{G}_5$, and $\mathbf{G}_6$ via (18), and reshape them to $\mathcal{G}_4 \in \mathbb{R}^{R_2\times R\times R_2}$, $\mathcal{G}_5 \in \mathbb{R}^{R_2\times B\times R_2}$, and $\mathcal{G}_6 \in \mathbb{R}^{R_2\times T\times R_2}$.
  Update $\mathcal{X}$ via (20).
  Refine mask $\mathcal{M}$ via (21) if $i \leq 5$.
  if $\|\mathcal{X} - \hat{\mathcal{X}}\|_F/\|\hat{\mathcal{X}}\|_F < 10^{-4}$ **then**
    Break the loop.
  end if
end for

Ensure: Reconstructed MTRSI $\mathcal{X} \in \mathbb{R}^{H\times W\times B\times T}$.

We present the pseudocode of the developed PAM-based algorithm for the proposed model in Algorithm 1. In general, each iteration of the algorithm incurs a computational complexity of $\mathcal{O}(H^4R + H^2R^4)$, under the assumption that $H = W \geq B = T$.

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The following Theorem provides a theoretical convergence guarantee for the developed algorithm.

Theorem 2 (Convergence Analysis): Assuming that the sequence $\{\mathcal{A}^{(s)}, \mathcal{B}^{(s)}, \mathcal{G}^{(s)}, \mathcal{X}^{(s)}\}_{s\in\mathbb{N}}$ generated by Algorithm 1 is bounded, it converges to a critical point of problem (5).

Proof. We reformulate the objective function of (5) as

$$f(\mathcal{A}, \mathcal{B}, \mathcal{G}, \mathcal{X}) = h(\mathcal{A}, \mathcal{B}, \mathcal{G}, \mathcal{X}) + g(\mathcal{X})$$

where $h(\mathcal{A}, \mathcal{B}, \mathcal{G}, \mathcal{X}) = (\alpha/2)\|\mathcal{X} - \mathcal{A} \times_3^1 \mathcal{B}\|_F^2 + (\beta_1/2)\|\mathcal{A} - \text{TR}(\{\mathcal{G}_k\}_{k=1}^3)\|_F^2 + (\beta_2/2)\|\mathcal{B} - \text{TR}(\{\mathcal{G}_k\}_{k=4}^6)\|_F^2$ and $g(\mathcal{X}) = (1/2)\|\mathcal{M} \odot (\mathcal{Y} - \mathcal{X})\|_F^2$.

According to [64], to prove Theorem 2, it suffices to verify that conditions 1)–4) are satisfied.

1) $f(\mathcal{A}, \mathcal{B}, \mathcal{G}, \mathcal{X})$ satisfies the Kurdyka–Łojasiewicz (K–Ł) property [65] at $\{\mathcal{A}^{(s)}, \mathcal{B}^{(s)}, \mathcal{G}^{(s)}, \mathcal{X}^{(s)}\}_{s\in\mathbb{N}}$.
2) $h(\mathcal{A}, \mathcal{B}, \mathcal{G}, \mathcal{X})$ is continuously differentiable (C^1) with a Lipschitz continuous gradient, and $g(\mathcal{X})$ is a proper lower semi-continuous function.
3) *Sufficient Decrease Condition:* For each $s \in \mathbb{N}$, there exists a constant $a \in (0, +\infty)$ such that

$$\begin{aligned}
&f\left(\mathcal{A}^{(s+1)}, \mathcal{B}^{(s)}, \mathcal{G}^{(s)}, \mathcal{X}^{(s)}\right) + a\|\mathcal{A}^{(s+1)} - \mathcal{A}^{(s)}\|_F^2 \\
&\le f\left(\mathcal{A}^{(s)}, \mathcal{B}^{(s)}, \mathcal{G}^{(s)}, \mathcal{X}^{(s)}\right) \\
&f\left(\mathcal{A}^{(s+1)}, \mathcal{B}^{(s+1)}, \mathcal{G}^{(s)}, \mathcal{X}^{(s)}\right) + a\|\mathcal{B}^{(s+1)} - \mathcal{B}^{(s)}\|_F^2 \\
&\le f\left(\mathcal{A}^{(s+1)}, \mathcal{B}^{(s)}, \mathcal{G}^{(s)}, \mathcal{X}^{(s)}\right) \\
&f\left(\mathcal{A}^{(s+1)}, \mathcal{B}^{(s+1)}, \mathcal{G}_{1:k}^{(s+1)}, \mathcal{G}_{k+1:6}^{(s)}, \mathcal{X}^{(s)}\right) \\
&\quad + a\|\mathcal{G}_k^{(s+1)} - \mathcal{G}_k^{(s)}\|_F^2 \\
&\le f\left(\mathcal{A}^{(s+1)}, \mathcal{B}^{(s+1)}, \mathcal{G}_{1:k-1}^{(s+1)}, \mathcal{G}_{k:6}^{(s)}, \mathcal{X}^{(s)}\right) \\
&f\left(\mathcal{A}^{(s+1)}, \mathcal{B}^{(s+1)}, \mathcal{G}^{(s+1)}, \mathcal{X}^{(s+1)}\right) \\
&\quad + a\|\mathcal{X}^{(s+1)} - \mathcal{X}^{(s)}\|_F^2 \\
&\le f\left(\mathcal{A}^{(s+1)}, \mathcal{B}^{(s+1)}, \mathcal{G}^{(s+1)}, \mathcal{X}^{(s)}\right)
\end{aligned}$$

where $k = 1, 2, \ldots, 6$ and $\mathcal{G}_{1:k} = \{\mathcal{G}_1, \mathcal{G}_2, \ldots, \mathcal{G}_k\}$.
4) *Relative Error Condition:* For each $s \in \mathbb{N}$, there exists

$$\begin{aligned}
&\mathcal{P}^{(s+1)} \in \partial_{\mathcal{A}} f\left(\mathcal{A}^{(s+1)}, \mathcal{B}^{(s)}, \mathcal{G}^{(s)}, \mathcal{X}^{(s)}\right) \\
&\mathcal{Q}^{(s+1)} \in \partial_{\mathcal{B}} f\left(\mathcal{A}^{(s+1)}, \mathcal{B}^{(s+1)}, \mathcal{G}^{(s)}, \mathcal{X}^{(s)}\right) \\
&\mathcal{U}_k^{(s+1)} \in \partial_{\mathcal{G}_k} f\left(\mathcal{A}^{(s+1)}, \mathcal{B}^{(s+1)}, \mathcal{G}_{1:k}^{(s+1)}, \mathcal{G}_{k+1:6}^{(s)}, \mathcal{X}^{(s)}\right) \\
&\mathcal{V}^{(s+1)} \in \partial_{\mathcal{X}} f\left(\mathcal{A}^{(s+1)}, \mathcal{B}^{(s+1)}, \mathcal{G}^{(s+1)}, \mathcal{X}^{(s+1)}\right)
\end{aligned}$$

and a constant $b \in (0, +\infty)$ such that

$$\begin{aligned}
\|\mathcal{P}^{(s+1)})\|_F &\le b\|\mathcal{A}^{(s+1)} - \mathcal{A}^{(s)}\|_F \\
\|\mathcal{Q}^{(s+1)})\|_F &\le b\|\mathcal{B}^{(s+1)} - \mathcal{B}^{(s)}\|_F \\
\|\mathcal{U}_k^{(s+1)})\|_F &\le b\|\mathcal{G}_k^{(s+1)} - \mathcal{G}_k^{(s)}\|_F \\
\|\mathcal{V}^{(s+1)}\|_F &\le b\|\mathcal{X}^{(s+1)} - \mathcal{X}^{(s)}\|_F
\end{aligned}$$

where $k = 1, 2, \ldots, 6$.

Since the objective function is a sum of several squared Frobenius norm terms, and each subproblem is differentiable, conditions 1)–4) are readily satisfied. The detailed proofs are omitted here; interested readers are encouraged to refer to [63] for more information. It is worth noting that the update of $\mathcal{M}$ is not included in the convergence analysis. This is because $\mathcal{M}$ is updated only a few times and thus has no effect on the overall convergence.

IV. Numerical Experiments

In this section, we conduct numerical experiments to evaluate the performance of the BTR decomposition-based method for MTRSI thick cloud removal. The main goal is to validate the following three claims.

- A: The BTR decomposition-based method can achieve superior cloud removal results compared to existing methods, while significantly reducing time costs, under the condition of an accurate cloud mask (Section IV-A).
- B: The BTR decomposition-based method can achieve superior cloud removal results compared to existing methods, while significantly reducing time costs, even when the cloud mask is inaccurate. (Section IV-B).
- C: The BTR decomposition-based method can achieve superior cloud removal results compared to existing methods, while significantly reducing time costs, under real-world cloud-covered scenarios. (Section IV-C).

We compare six methods for MTRSI thick cloud removal, including mask-unknown methods: TVLRSDC [32] and BC-SLRpGS [33]; mask-known methods: TRTV [31], TW [34], and STDTV [36]; and a mask-semi-known method: SSTC-CR [35]. Among them, TVLRSDC relies on matrix decomposition, BC-SLRpGS utilizes tensor SVD, TRTV adopts TR decomposition, TW implements tensor wheel decomposition, STDTV employs stacked Tucker decomposition, and SSTC-CR is based on FCTN decomposition. The hyperparameters of all compared methods are manually adjusted based on the recommendations in the corresponding papers and codes to achieve their optimal performance. All test data are normalized into [0, 1]. All experiments are conducted on a laptop equipped with Windows 11, a 13th Gen Intel[2] Core i7-13650HX processor (2.60 GHz), 24 GB of RAM, and an NVIDIA GeForce RTX 4060 GPU. STDTV [36] is executed using GPU on PyTorch 2.3.0, while all other methods (including ours) are executed using CPU on MATLAB R2022a.

A. Simulated Experiments With Accurate Masks

We conduct experiments to validate Claim A, which considers the condition of accurate cloud masks.

1) Experimental Setting: We utilize 20 MTRSIs collected from five regions: *Africa*, *America*, *Eastern Asia*, *Western Asia*, and *Europe*, as test simulated data [4]. Each region consists of four images, as shown in Fig. 2. All test data have a spatial resolution of 256 × 256 pixels, spanning 13 spectral bands and 6 temporal nodes. The test data cover both scenarios with significant temporal variations and scenarios with relatively minor changes over time. We assume that the spatial position of clouds across all spectral bands is consistent at the same temporal node and that clouds exist in the first three temporal nodes. We employ two types of masks, as shown in Table I. One is a simulated mask used in the related work [35] and the other is a mask extracted from real cloud scenarios.[3] In particular, the simulated mask is randomly introduced into the first three temporal nodes, with its spatial location also selected

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[3]https://dataspace.copernicus.eu/

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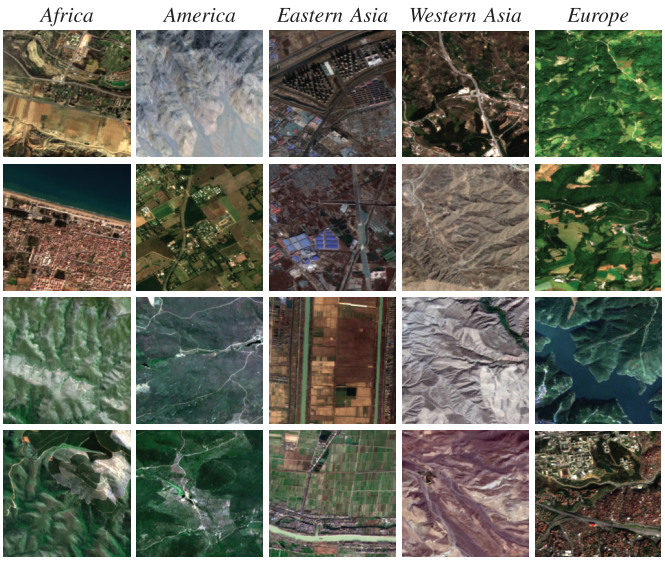
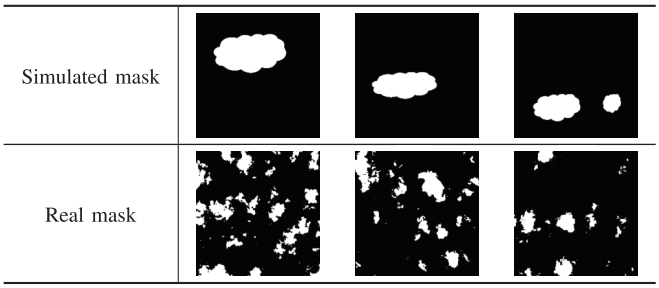


Fig. 2. MTRSIs (shown in RGB) used for simulated experiments [4].

TABLE I

MASKS USED FOR SIMULATED EXPERIMENTS WITH ACCURATE MASKS. THE CLOUD AREAS ARE DISPLAYED IN WHITE



randomly. We employ two metrics to quantitatively evaluate the cloud removal performance, including peak signal-to-noise ratio (PSNR) and structure similarity index measure (SSIM). Their definitions are as follows:

$$\mathrm{PSNR} = \frac{1}{B|\mathbb{T}|}\sum_{b=1}^{B}\sum_{t\in\mathbb{T}} 10\log_{10}\frac{HW}{\|\mathbf{X}_b^t - \hat{\mathbf{X}}_b^t\|_F^2}$$

and

$$\mathrm{SSIM} = \frac{1}{B|\mathbb{T}|}\sum_{b=1}^{B}\sum_{t\in\mathbb{T}} \frac{\left(2\nu_{\mathbf{X}_b^t}\nu_{\hat{\mathbf{X}}_b^t}+c_1\right)\left(2\omega_{\mathbf{X}_b^t,\hat{\mathbf{X}}_b^t}+c_2\right)}{\left(\nu_{\mathbf{X}_b^t}^2+\nu_{\hat{\mathbf{X}}_b^t}^2+c_1\right)\left(\omega_{\mathbf{X}_b^t}^2+\omega_{\hat{\mathbf{X}}_b^t}^2+c_2\right)}$$

where $\mathbb{T}$ denotes the index set of cloud-contaminated temporal nodes; $|\mathbb{T}|$ denotes the number of elements in $\mathbb{T}$; $\mathbf{X}_b^t$ and $\hat{\mathbf{X}}_b^t$ are the clean and reconstructed images at the bth spectral band and the tth temporal node, respectively; $\nu_{\mathbf{X}_b^t}$ is the mean value of $\mathbf{X}_b^t$; $\omega_{\mathbf{X}_b^t,\hat{\mathbf{X}}_b^t}$ is the covariance between $\mathbf{X}_b^t$ and $\hat{\mathbf{X}}_b^t$; $\omega_{\mathbf{X}_b^t}$ is the standard deviation of $\mathbf{X}_b^t$; and c_1 and c_2 are constants. In all tests, we set $R = 40, R_1 = 4, R_2 = 2$, and $\beta_1 = \beta_2 = 1$.

2) Result Analysis: In Tables II and III, we report the PSNR, SSIM, and running time for different MTRSI cloud removal methods using an accurate simulated cloud mask and an accurate real cloud mask, respectively. It is evident that our method consistently outperforms the comparison methods in two key aspects: reconstruction quality and computational efficiency. In terms of reconstruction quality, our method achieves superior PSNR and SSIM values across most cases. Notably, for the simulated mask, the PSNR value of our method exceeds that of the second-best method by 2–3 dB, highlighting a significant improvement in reconstruction quality. The most remarkable advantage is in computational efficiency. Our method completes the task under 10 s, whereas all comparison methods require over 100 s, with several approaching 1000 s, that is, our method achieves a speedup of 10–100 times. This advantage can be attributed to two aspects. In theory, BTR decomposition inherently enjoys relatively low computational complexity. In practice, BTR decomposition is tailored to the unique characteristics of MTRSIs. This enables us to use very small rank values to achieve satisfactory performance. The detailed rationale can be found in Remarks 1 and 2 and Section IV-D1.

To further facilitate visual comparison, Fig. 3 presents the reconstructed MTRSIs (shown in RGB) obtained by different MTRSI cloud removal methods using an accurate simulated cloud mask, showcasing results from images in *Africa*, *Eastern Asia*, and *Europe* at specific temporal nodes. Similarly, Fig. 4 displays the reconstructed MTRSIs (shown in RGB) obtained by different MTRSI cloud removal methods using an accurate real cloud mask, showcasing results from the second image in *America* across three temporal nodes. We observe that our method consistently delivers superior visual results compared to the comparison methods, particularly excelling in the reconstruction of intricate shape structures and fine texture details. Specifically, while the comparison methods are able to reconstruct the overall structure of the image, their results contain evident blurry areas. In contrast, our method effectively reconstructs finer details and shapes, such as the "field paths" in the second row of Fig. 3 and the "stadium" in the first row of Fig. 4.

B. Simulated Experiments With Inaccurate Masks

We conduct experiments to validate Claim B, which considers the condition of inaccurate cloud masks.

1) Experimental Setting: We also employ the MTRSIs used in Section IV-A as the test data. For each test MTRSI, we use the guidance mask from Table IV as the initial mask for the different MTRSI cloud removal methods. However, the actual cloud mask introduced to the first three temporal nodes of the MTRSI is randomly selected from the three cases listed in Table IV, meaning that the initial mask is not accurate. The parameter settings are consistent with those in Section IV-A.

2) Result Analysis: Table V lists the PSNR, SSIM, and running time for different MTRSI cloud removal methods using inaccurate simulated cloud masks. It can be seen that our method consistently surpasses the comparison methods in terms of PSNR and SSIM across all datasets while achieving a speedup of 10–100 times. In Fig. 5, we present the reconstructed MTRSIs (shown in RGB) obtained by different cloud removal methods using an accurate simulated cloud mask, showcasing results from images in *America*, *Eastern Asia*, and

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TABLE II

COMPARISON OF PSNR (MEAN ± STANDARD DEVIATION, ↑), SSIM (MEAN ± STANDARD DEVIATION, ↑), AND RUNNING TIME (MEAN ± STANDARD DEVIATION, ↓) FOR DIFFERENT MTRSI CLOUD REMOVAL METHODS WITH AN ACCURATE SIMULATED CLOUD MASK, WHERE THE MEAN AND STANDARD DEVIATION ARE CALCULATED FROM THE RESULTS OF FOUR TEST MTRSIS IN EACH REGION

Region	Index	TRTV (*TGRS'19*)	TVLRSDC (*ISPRS'19*)	BC-SLRpGS (*TGRS'22*)	TW (*NeurIPS'22*)	SSTC-CR (*ISPRS'23*)	STDTV (*TGRS'24*)	Ours
Africa	PSNR	35.581±2.215	34.737±3.303	34.651±3.568	35.918±3.249	37.705±3.111	37.492±3.895	**39.839±4.353**
	SSIM	0.9662±0.009	0.9704±0.011	0.9682±0.011	0.9685±0.012	0.9763±0.009	0.9733±0.011	**0.9822±0.008**
	Time (s)	145.78±5.935	268.73±0.525	186.65±35.92	842.83±6.643	343.65±64.66	232.10±0.816	**8.9386±1.561**
America	PSNR	39.337±2.489	38.029±2.796	37.002±1.998	39.815±2.903	39.963±1.848	41.093±2.381	**44.563±4.393**
	SSIM	0.9748±0.009	0.9800±0.009	0.9773±0.007	0.9782±0.009	0.9827±0.007	0.9740±0.004	**0.9893±0.005**
	Time (s)	142.61±1.209	268.74±0.331	170.69±29.13	835.46±28.99	248.73±136.3	232.37±0.644	**6.8660±2.408**
Eastern Asia	PSNR	34.168±0.696	32.003±1.019	32.649±2.255	33.946±0.778	35.767±0.703	35.225±0.763	**37.594±0.850**
	SSIM	0.9645±0.006	0.9611±0.009	0.9639±0.012	0.9657±0.007	0.9728±0.006	0.9693±0.004	**0.9790±0.004**
	Time (s)	141.85±0.239	268.61±0.312	145.10±40.34	767.07±46.29	368.59±67.13	232.75±1.608	**9.0194±0.719**
Western Asia	PSNR	36.127±1.637	34.371±1.203	35.334±2.344	36.070±2.213	37.870±2.159	37.785±2.058	**40.079±2.301**
	SSIM	0.9676±0.007	0.9730±0.006	0.9734±0.007	0.9697±0.009	0.9805±0.008	0.9743±0.006	**0.9844±0.004**
	Time (s)	141.78±0.549	269.04±0.325	180.73±39.80	792.24±67.49	326.15±69.60	232.09±0.867	**7.9467±1.739**
Europe	PSNR	34.414±1.010	32.643±1.298	32.149±2.919	33.885±1.391	35.743±1.247	36.295±1.338	**37.723±1.459**
	SSIM	0.9618±0.004	0.9653±0.005	0.9639±0.007	0.9597±0.008	0.9702±0.004	0.9703±0.002	**0.9777±0.003**
	Time (s)	141.82±0.885	268.91±0.159	165.47±38.16	802.3±44.206	358.29±28.36	232.35±0.366	**9.5036±0.233**

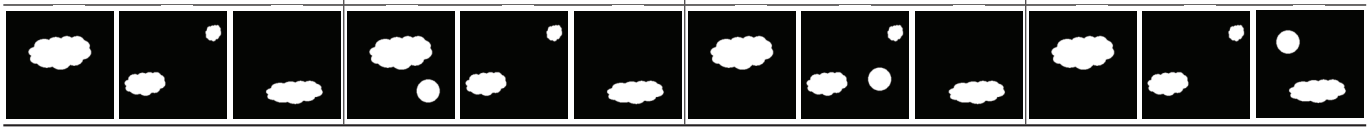
TABLE III

COMPARISON OF PSNR (MEAN ± STANDARD DEVIATION, ↑), SSIM (MEAN ± STANDARD DEVIATION, ↑), AND RUNNING TIME (MEAN ± STANDARD DEVIATION, ↓) FOR DIFFERENT MTRSI CLOUD REMOVAL METHODS WITH AN ACCURATE REAL CLOUD MASK, WHERE THE MEAN AND STANDARD DEVIATION ARE CALCULATED FROM THE RESULTS OF FOUR TEST MTRSIS IN EACH REGION

Region	Index	TRTV (*TGRS'19*)	TVLRSDC (*ISPRS'19*)	BC-SLRpGS (*TGRS'22*)	TW (*NeurIPS'22*)	SSTC-CR (*ISPRS'23*)	STDTV (*TGRS'24*)	Ours
Africa	PSNR	35.474±1.242	29.820±1.216	33.540±2.001	35.794±2.353	37.405±2.040	38.233±2.189	**39.301±2.938**
	SSIM	0.9586±0.009	0.9407±0.012	0.9512±0.014	0.9614±0.015	0.9700±0.011	0.9730±0.011	**0.9755±0.010**
	Time (s)	148.56±7.278	270.60±1.748	150.51±15.01	795.84±59.37	412.09±55.12	233.75±1.957	**8.8082±1.777**
America	PSNR	38.026±2.015	32.727±1.233	35.089±1.185	39.027±2.651	39.043±1.615	40.128±1.873	**42.651±2.961**
	SSIM	0.9700±0.009	0.9563±0.011	0.9649±0.009	0.9754±0.010	0.9780±0.006	0.9790±0.005	**0.9861±0.006**
	Time (s)	144.83±1.056	270.90±1.020	154.53±37.47	824.47±32.50	291.65±158.1	231.21±0.291	**6.8704±2.461**
Eastern Asia	PSNR	34.715±1.029	29.840±1.220	31.523±2.002	34.704±1.298	35.781±1.385	36.395±1.050	**36.613±1.332**
	SSIM	0.9571±0.005	0.9348±0.011	0.9444±0.012	0.9602±0.006	0.9670±0.004	0.9683±0.003	**0.9705±0.003**
	Time (s)	145.45±2.035	270.02±0.648	128.78±26.87	845.18±17.42	421.17±49.05	231.51±1.130	**9.3418±0.073**
Western Asia	PSNR	35.215±1.312	31.000±1.136	33.345±2.068	35.518±1.718	36.678±2.200	37.328±1.930	**38.650±3.038**
	SSIM	0.9606±0.006	0.9468±0.009	0.9561±0.009	0.9648±0.008	0.9710±0.007	0.9730±0.005	**0.9774±0.007**
	Time (s)	143.56±0.058	270.17±0.724	132.28±16.72	830.18±25.22	387.76±32.24	231.45±0.570	**8.9721±0.736**
Europe	PSNR	34.429±0.796	29.373±1.604	31.053±2.904	34.377±1.036	35.459±1.138	**36.490±0.970**	36.480±1.612
	SSIM	0.9539±0.004	0.9320±0.011	0.9446±0.009	0.9557±0.005	0.9630±0.004	**0.9688±0.002**	0.9673±0.007
	Time (s)	144.87±2.506	270.17±0.162	127.43±31.53	804.81±26.77	437.60±30.16	231.87±0.446	**9.3607±0.038**

TABLE IV

GUIDANCE AND ACTUAL MASKS USED FOR SIMULATED EXPERIMENTS WITH INACCURATE MASKS. THE CLOUD AREAS ARE DISPLAYED IN WHITE

Guidance mask	Actual mask 1	Actual mask 2	Actual mask 3
			

Europe at specific temporal nodes. We observe that the mask-known methods TRTV, TW, and STDTV fail to remove clouds beyond the scope of the initial mask, as they do not account for mask update throughout the cloud removal process. In contrast, mask-unknown and semi-known methods do not rely solely on the initial mask, allowing them to effectively remove clouds beyond the initial scope. In particular, our method delivers the best visual results among all utilized methods, demonstrating exceptional performance even in scenarios with complex spatial structures, such as the scene in the third row of Fig. 5. This demonstrates that our method is not only robust to inaccuracies in the initial cloud masks but also excels at capturing intricate spatial patterns and adapting to complex scenarios, all while achieving significantly enhanced processing speeds.

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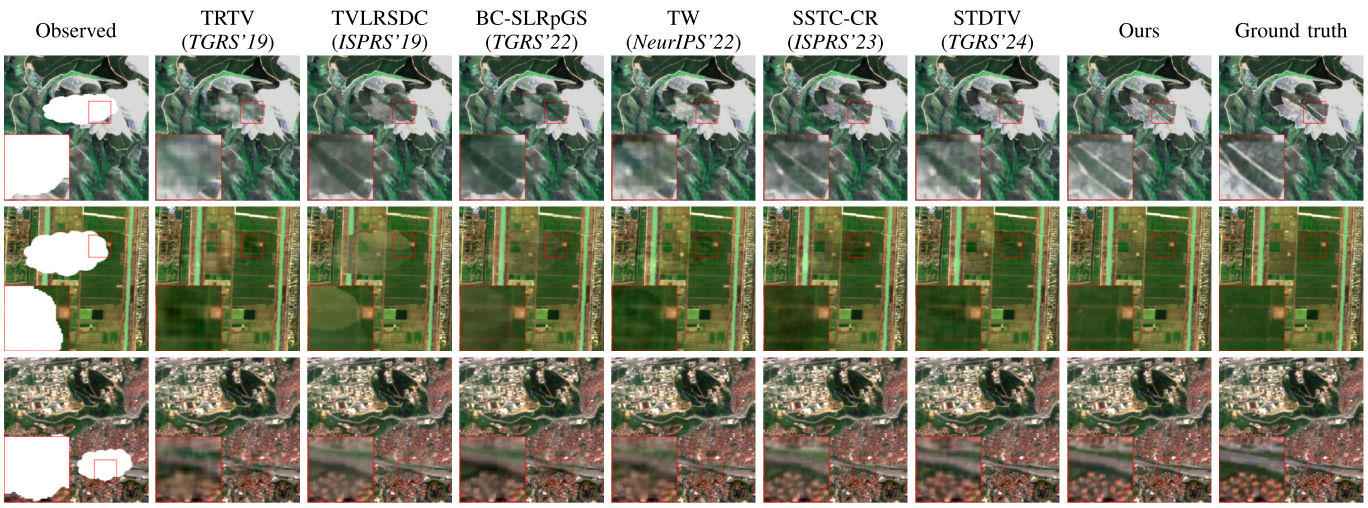


Fig. 3. Reconstructed MTRSIs (shown in RGB) obtained by different methods with an accurate simulated cloud mask. Results of the fourth image in *Africa* at temporal node 1, the third image in *Eastern Asia* at temporal node 1, and the fourth image in *Europe* at temporal node 3 are from top to bottom.

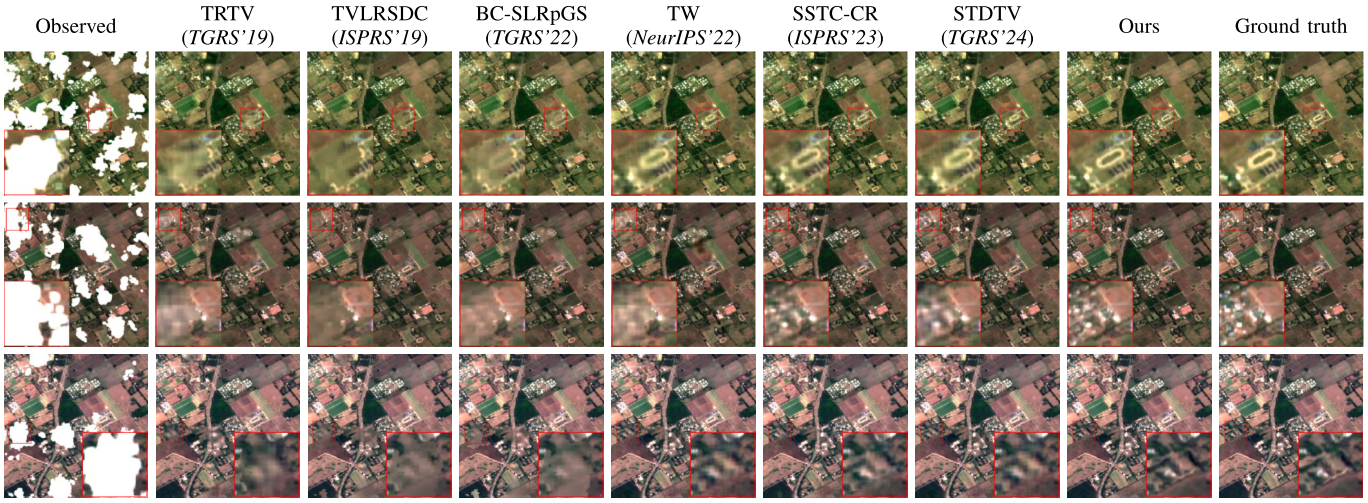


Fig. 4. Reconstructed MTRSIs (shown in RGB) obtained by different methods with an accurate real cloud mask. Results of the second image in *America* at temporal nodes 1–3 are from top to bottom.

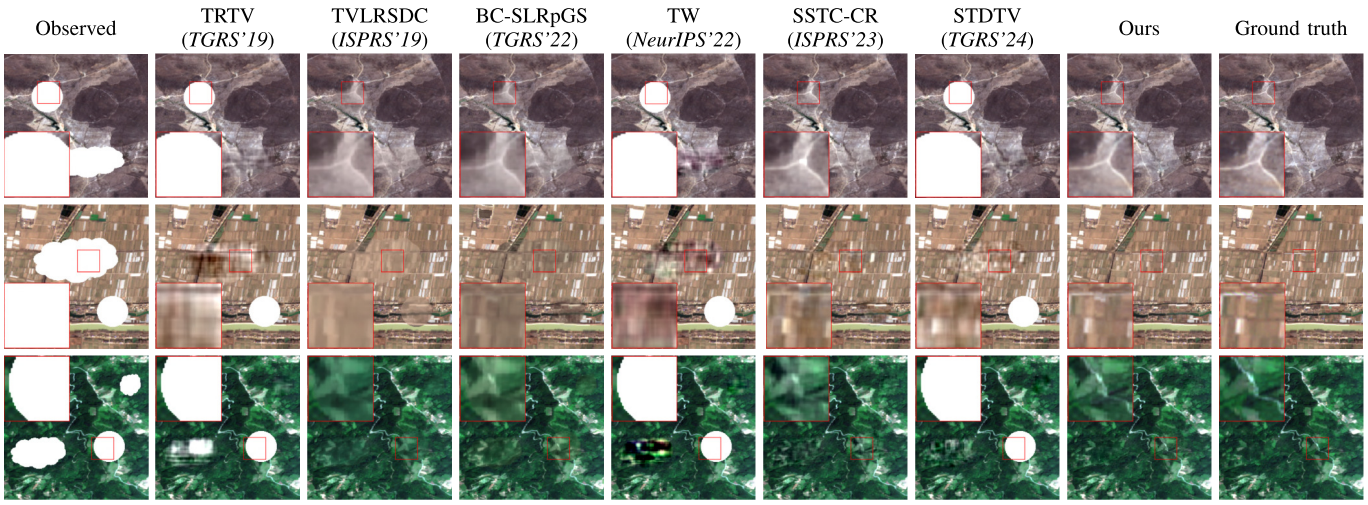


Fig. 5. Reconstructed MTRSIs (shown in RGB) obtained by different methods with inaccurate simulated cloud masks. Results of the fourth image in *America* at temporal node 3, the fourth image in *Eastern Asia* at temporal node 1, and the first image in *Europe* at temporal node 2 are from top to bottom.

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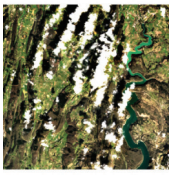
TABLE V

COMPARISON OF PSNR (MEAN ± STANDARD DEVIATION, ↑), SSIM (MEAN ± STANDARD DEVIATION, ↑), AND RUNNING TIME (MEAN ± STANDARD DEVIATION, ↓) FOR DIFFERENT MTRSI CLOUD REMOVAL METHODS WITH INACCURATE SIMULATED CLOUD MASKS, WHERE THE MEAN AND STANDARD DEVIATION ARE CALCULATED FROM THE RESULTS OF FOUR TEST MTRSIS IN EACH REGION

Region	Index	TRTV (*TGRS'19*)	TVLRSDC (*ISPRS'19*)	BC-SLRpGS (*TGRS'22*)	TW (*NeurIPS'22*)	SSTC-CR (*ISPRS'23*)	STDTV (*TGRS'24*)	Ours
Africa	PSNR	29.849±1.274	34.115±3.083	34.190±3.392	29.539±1.792	36.645±3.389	31.320±1.386	**38.835±3.735**
	SSIM	0.9462±0.011	0.9660±0.012	0.9649±0.011	0.9452±0.014	0.9700±0.010	0.9560±0.010	**0.9795±0.008**
	Time (s)	144.93±0.718	272.09±0.621	185.57±34.85	681.75±14.95	328.03±42.97	242.88±1.301	**8.6685±1.133**
America	PSNR	32.478±2.233	37.353±2.761	36.558±1.978	32.827±2.427	38.779±1.942	33.907±1.649	**43.233±3.747**
	SSIM	0.9568±0.011	0.9765±0.011	0.9748±0.008	0.9573±0.013	0.9778±0.006	0.9643±0.007	**0.9876±0.006**
	Time (s)	144.87±1.131	271.31±0.472	173.19±40.02	663.92±33.37	252.19±130.3	241.87±0.852	**7.4045±2.802**
Eastern Asia	PSNR	27.943±0.933	31.225±1.306	32.068±2.131	27.519±0.730	35.349±1.107	29.075±0.465	**36.682±0.804**
	SSIM	0.9432±0.004	0.9571±0.008	0.9608±0.010	0.9415±0.006	0.9710±0.004	0.9521±0.002	**0.9777±0.002**
	Time (s)	145.18±1.711	271.67±0.128	148.47±40.53	703.44±44.26	391.24±59.55	241.96±0.359	**9.2601±0.348**
Western Asia	PSNR	29.618±2.131	33.293±1.289	34.486±1.974	29.446±2.303	36.584±2.113	31.060±1.727	**38.367±3.247**
	SSIM	0.9495±0.010	0.9683±0.006	0.9688±0.008	0.9503±0.011	0.9739±0.008	0.9583±0.006	**0.9806±0.007**
	Time (s)	144.74±0.157	271.64±0.313	174.61±29.89	857.64±28.40	266.75±72.82	242.37±0.205	**8.6801±1.007**
Europe	PSNR	29.069±1.577	32.894±1.761	31.973±2.746	28.816±1.121	35.508±1.100	30.625±0.518	**37.979±1.015**
	SSIM	0.9389±0.010	0.9606±0.009	0.9596±0.007	0.9344±0.014	0.9648±0.006	0.9518±0.005	**0.9744±0.005**
	Time (s)	145.13±2.769	271.36±0.515	156.96±28.50	759.30±47.99	412.01±27.46	242.26±0.577	**9.5372±0.036**

TABLE VI

MTRSIS (SHOWN IN RGB) USED FOR REAL EXPERIMENTS

Bourgogne (size: 600 × 600 × 7 × 4)				*Provence* (size: 1000 × 1000 × 7 × 4)			
02/24/2020	02/08/2020	10/03/2019	09/17/2019	07/29/2021	07/13/2021	06/11/2021	05/26/2021
							

C. Experiments With Real-World Cloud-Covered Scenarios

We conduct experiments to validate Claim C, which considers real-world cloud-covered scenarios.

1) Experimental Setting: We employ two MTRSIs shown in Table VI as test data, named *Bourgogne* and *Provence*, which were captured over Bourgogne and Provence, France, by the Landsat-8 satellite sensor with a ground sampling distance of 30 m.[4] The corresponding masks are also acquired from the website.[4] In all tests, we set $R = 20, R_1 = 6, R_2 = 2$, and $\beta_1 = \beta_2 = 10$.

2) Result Analysis: Fig. 6 displays the reconstructed MTRSIs (shown in RGB) obtained by different cloud removal methods for *Bourgogne* at temporal node 1 and *Provence* at temporal node 2, along with the corresponding running times. It can be observed that our method delivers superior reconstruction performance, especially excelling in the preservation of local details and the restoration of texture patterns. In contrast, the results from TRTV, TVLRSDC, and BC-SLRpGS exhibit noticeable information ambiguity, while the results from TW contain noise artifacts. SSTC-CR and STDTV perform reasonably well on *Bourgogne*, but the results on *Provence* still suffer from noise interference or texture loss. Notably, our method not only delivers superior reconstruction results but also outperforms the comparison methods in terms of running time, achieving a speed improvement of 10–100 times. The results also highlight the efficiency of our method in handling large-scale data, with the potential for real-time processing on more advanced platforms.

D. Discussion

1) Parameter Analysis: We analyze the influence of parameters on the performance of our method. Specifically, the main parameters that require adjustment in our method are R, R_1, R_2, β_1, and β_2. Among them, R, R_1, and R_2 represent the ranks in BTR decomposition, while β_1 and β_2 are penalty parameters. For simplicity, we set $\beta_1 = \beta_2$ and denote them collectively as β in all tests. We adopt the experimental setting in Section IV-A.

Figs. 7– 9 present the average PSNR values obtained by our method on four MTRSIs from regions *America* and *Western Asia*, with respect to different R, R_1, and R_2, respectively. It can be observed that the performance of our method remains relatively stable when $R \geq 40$ and is not particularly sensitive to the choice of R_1. In contrast, R_2 shows a more noticeable influence, with better performance achieved when $R_2 \leq 4$. Therefore, we recommend choosing R from the candidate $\{20, 30, 40, 50, 60\}, R_1$ from the candidate set $\{2, 4, 6, 8\}$, and R_2 from $\{2, 3, 4\}$.

Fig. 10 presents the average PSNR values obtained by our method on four MTRSIs from regions *America* and *Western*

[4]https://theia.cnes.fr/atdistrib/rocket/#/home

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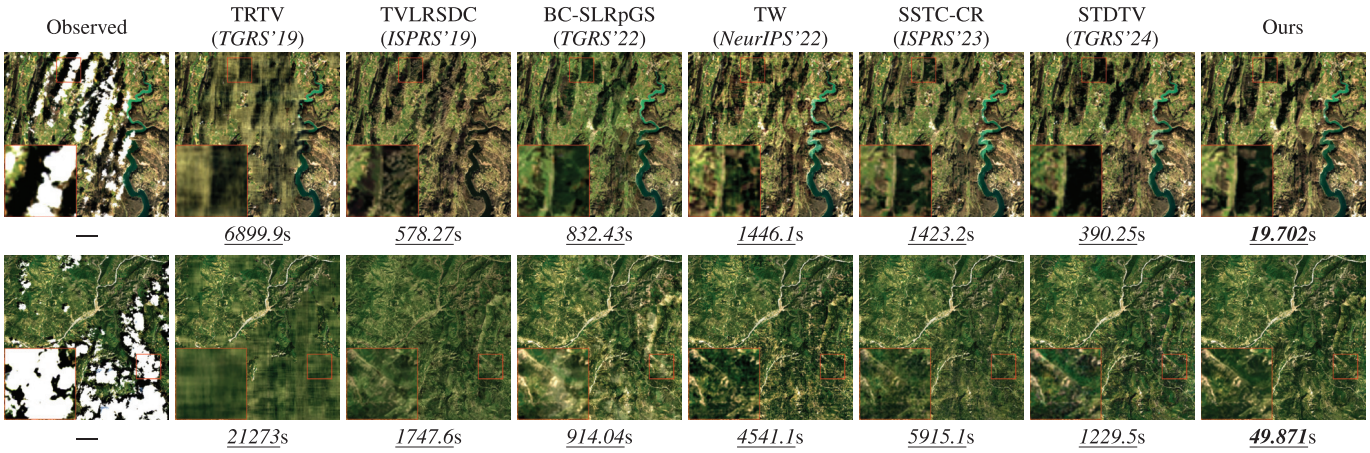


Fig. 6. Reconstructed MTRSIs (shown in RGB) obtained by different methods with real-world cloud-covered scenarios. The first and second rows are the results of MTRSI *Bourgogne* at temporal node 1 and MTRSI *Provence* at temporal node 2, respectively.

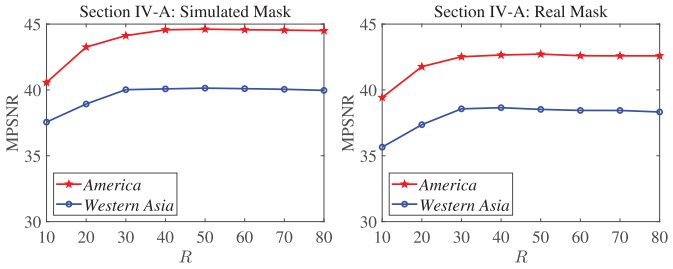


Fig. 7. Influence of the parameter R on the performance of our method.

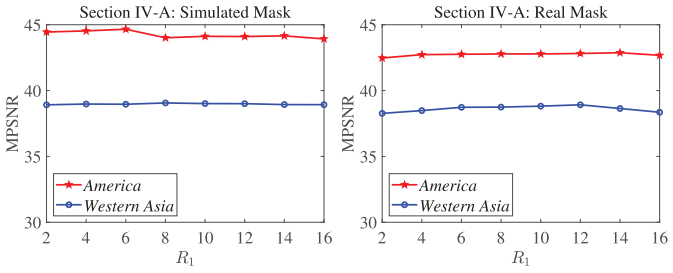


Fig. 8. Influence of the parameter R_1 on the performance of our method.

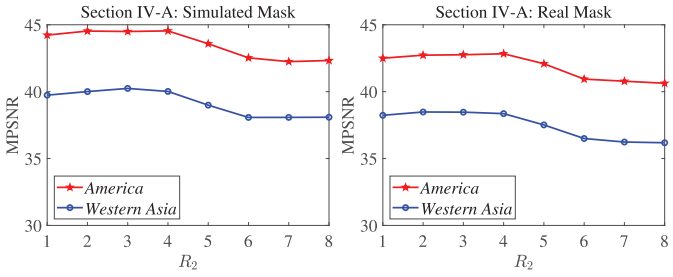


Fig. 9. Influence of the parameter R_2 on the performance of our method.

Asia with respect to different β's. It can be seen that the performance of our method is slightly sensitive to β. We suggest selecting β from $\{0.1, 0.5, 1, 5, 10\}$.

2) Numerical Convergence Analysis: We conduct a numerical convergence analysis to further support the theoretical convergence result. Fig. 11 presents the relative change

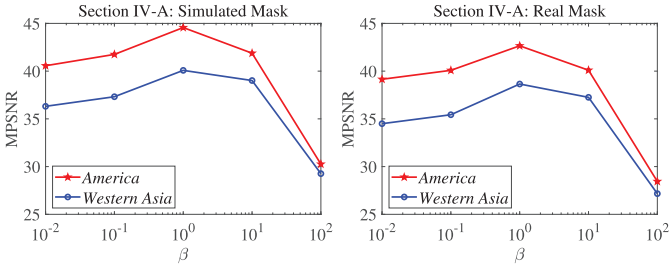


Fig. 10. Influence of the parameter β on the performance of our method.

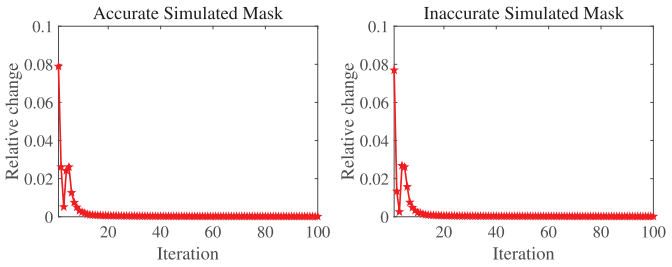


Fig. 11. Relative change curves with respect to the iteration number under both accurate and inaccurate simulated masks on the second image in *America*.

between the reconstructed MTRSI in each iteration and that in the previous iteration, under both accurate and inaccurate simulated masks, using the second image in *America*. It can be observed that the relative change obtained by the developed algorithm decreases monotonically and gradually converges to zero as the number of iterations increases, demonstrating the algorithm's numerical convergence.

3) Mask Robustness Analysis: We evaluate the robustness of our method under inaccurate simulated cloud masks with varying false negative rates (FNRs), where FNR is defined as the proportion of cloud pixels mislabeled as clear. Table VII presents the PSNR, SSIM, and running time of our method under different FNRs. As observed, although the PSNR and SSIM values obtained by our method slightly decrease with the increase in FNR, the performance degradation is marginal. This slight decline is mainly due to the increased area of cloud

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TABLE VII

PSNR (MEAN ± STANDARD DEVIATION, ↑), SSIM (MEAN ± STANDARD DEVIATION, ↑), AND RUNNING TIME (MEAN ± STANDARD DEVIATION, ↓) OF OUR METHOD UNDER INACCURATE SIMULATED CLOUD MASKS WITH DIFFERENT FNRS, WHERE THE MEAN AND STANDARD DEVIATION ARE CALCULATED FROM THE RESULTS OF FOUR TEST MTRSIS IN AMERICA

FNR	5%	15%	25%	35%
PSNR	42.834±4.034	42.311±4.234	41.793±4.457	41.321±4.585
SSIM	0.9881±0.006	0.9863±0.006	0.9845±0.007	0.9819±0.008
Time (s)	7.4419±2.535	7.5283±2.541	7.4088±2.684	7.7206±2.515

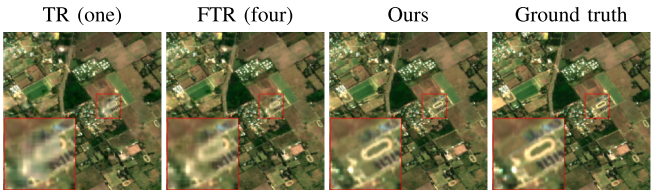


Fig. 12. Reconstructed MTRSIs (shown in RGB) of TR, FTR, and our method with an accurate real cloud mask on the second image in *America* at temporal node 1.

TABLE VIII

COMPARISON OF PSNR (MEAN ± STANDARD DEVIATION, ↑), SSIM (MEAN ± STANDARD DEVIATION, ↑), AND RUNNING TIME (MEAN ± STANDARD DEVIATION, ↓) FOR TR, FTR, AND OUR METHOD WITH AN ACCURATE SIMULATED CLOUD MASK, WHERE THE MEAN AND STANDARD DEVIATION ARE CALCULATED FROM THE RESULTS OF FOUR TEST MTRSIS IN AMERICA

Method	TR	FTR	Ours
PSNR	40.246±3.218	41.435±2.832	**44.563±4.393**
SSIM	0.9784±0.008	0.9813±0.007	**0.9893±0.005**
Time (s)	41.068±9.201	313.22±5.991	**6.8660±2.408**

coverage, which leads to more severe information loss in the input data. These results indicate that our method is robust to different levels of undetected cloud coverage.

4) Ablation Study: We conduct numerical experiments to compare the performance of standard TR decomposition (single ring), the method that adopts four ring structures, referred to as four-TR (FTR) decomposition,[5] and our BTR method (bilateral ring). Table VIII reports the PSNR, SSIM, and running time of TR, FTR, and our BTR method. It can be observed that the BTR method not only improves reconstruction quality (in terms of PSNR and SSIM) compared to TR and FTR but also achieves lower computational cost. Fig. 12 presents the reconstructed MTRSIs (shown in RGB) obtained by TR, FTR, and our BTR method using an accurate simulated cloud mask based on the second image in *America* at temporal node 1. We observe that the BTR method provides more accurate reconstruction results than TR and FTR, especially in preserving intricate structural shapes and fine texture details. The superior performance of our method is attributed to the ability of BTR decomposition to capture the unique correlation structure of MTRSIs (see Remark 1), particularly the weak correlation between the two spatial dimensions and the strong correlation between the spectral and temporal dimensions. This customized representation enables the use of very small rank values to achieve satisfactory cloud removal performance. In contrast, TR and FTR fail to fully exploit these correlation patterns, which leads to inferior reconstruction performance.

[5]FTR decomposition essentially performs a second-level TR decomposition on each individual factor of the standard TR decomposition.

V. CONCLUSION

In this article, we constructed a tensor decomposition specifically tailored for MTRSIs, called BTR decomposition, which enables a customized representation of spatial, spectral, and temporal correlations with reduced computational and storage complexity. Based on BTR decomposition, we proposed an MTRSI cloud removal model and developed a PAM-based algorithm to solve it with a convergence guarantee. The experimental results verified that our method achieved superior performance and provided a 10- to 100-fold improvement in computational efficiency compared to existing tensor-based methods.

In our method, there is no complex regularization; the constructed BTR decomposition alone is sufficient to achieve superior results compared to existing methods while significantly reducing computational time. Of course, it is precisely for this reason that the method's ability to reconstruct fine image details remains limited, as it leverages only the global correlations of MTRSIs without incorporating additional prior knowledge. In the future, researchers could investigate incorporating advanced regularization terms to capture additional prior knowledge, which would open up opportunities for further performance enhancement, potentially improving accuracy, robustness, and generalization in various applications. In addition, the automatic rank selection strategy is currently not included in our framework. While we have empirically demonstrated robustness to a range of rank settings, developing a data-driven or adaptive rank selection mechanism remains an important direction for future work.

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